

TANK TURRET ROTATION

Mohamed Thebti

October 12, 2025

Contents

1	Introduction	3
2	Coordinates	3
3	Computation algorithm	4
3.1	Following trigonometric circle	4
3.2	Following clockwise direction	4
4	Schéma cinématique	6
4.1	Vecteurs positions	6
5	Angular momentum	7
5.1	Formula	7
5.2	Condition of stability	7
5.3	Pivots à droite et à gauche	8
6	Conclusion	8

1 Introduction

The context is a rotating turret that can face any direction (360 freedom movement). The turret is ordered (manually or automatically) to face a certain direction. The objective of this report is to compute in which rotation direction the turret must rotate in order to minimize the time needed to reach the facing direction.

Example of application :

- Autonomous observation turret (on a tank, on scout vehicle, IFV, defensive position)
- Drone camera turret
- Tank turret

As a bonus, the same calculation will be done to estimate the gun elevation.

2 Coordinates

Let's consider the trigonometric circle as reference. The rotation is positive if it is following the trigonometric direction, which is anti-clockwise.

For the turret elevation, 0° is the horizon level. Angle is positive when the elevation is moving up, negative if turret facing below the horizon.

The turret is facing a specified angle, α_{start} , with β_{start} as elevation angle. a target is situated at an angle α_{finish} and β_{start} . The turret must rotate in order to satisfy the following conditions :

- Starting angle is equal to the finish angle

$$\alpha_{start} = \alpha_{finish} \quad (1)$$

- Elevation angle are equal

$$\beta_{start} = \beta_{finish} \quad (2)$$

- This movement must take the shortest time.

The calculation result will indicate in which direction the turret must rotate.

3 Computation algorithm

The angle are first converted to trigonometric range $[0^\circ, 360^\circ]$ ($[0, 2\pi]$)

To find the correct rotation direction, compute the difference of angles $\angle$ between the starting angle and the finish angle.

The turret elevation is restricted to the range $[-180^\circ, +180^\circ]$ ($[-\pi, +\pi]$). The sign of difference of angle will give the correct movement direction.

3.1 Following trigonometric circle

$$\Delta_{trigo} = \alpha_{finish} - \alpha_{start} \quad (3)$$

3.2 Following clockwise direction

The angle will be converted to the clockwise reference :

$$\alpha_{start}^{clockwise} = 360 - \alpha_{start} \quad (4)$$

$$\alpha_{start}^{clockwise} = 2\pi - \alpha_{start} \quad (5)$$

$$\alpha_{finish}^{clockwise} = 360 - \alpha_{finish} \quad (6)$$

$$\alpha_{finish}^{clockwise} = 2\pi - \alpha_{finish} \quad (7)$$

$$\Delta_{clockwise} = \alpha_{finish}^{clockwise} - \alpha_{start}^{clockwise} \quad (8)$$

The trigonometric circle is divided into 4 quadrants. When calculating the difference between angles, it is useful to know to which quadrant each angle belongs. In case of two neighbor quadrants, the difference is easily found. If the angles are in opposite positions (quadrant 1 and 3 or 2 and 4), there is two possibilities to compute the difference. Because we want to shortest path for the turret, the difference of angle should be :

$$\delta = \alpha_{finish} - \alpha_{start} \leq \pi \quad (9)$$

In case the result does not fulfill this condition, δ must be computed

table

Screw	Threaded hole diameter	Bore Hole (clearance)
M3	2.8	3.2
4	35	144
5	45	300

Clearance To fix two part together, a clearance is needed. When two parts must be assembled together, a clearance of 0.1mm is enough. Then use the glue to fix the assembly.

Case	Method#1	Method#2	Method#3
1	50	837	970
2	47	877	230
3	31	25	415
4	35	144	2356
5	45	300	556

$$Length_{latitude}(\phi) = 111132.92 - 559.82 \cdot \cos(2 \cdot \phi) + 1.175 \cdot \cos(4 \cdot \phi) - 0.0023 \cdot \cos(6 \cdot \phi) = \dots [m/degree] \quad (10)$$

1 degree longitude at latitude phi

$$Length_{longitude}(\phi) = 111412.84 - 93.5 \cdot \cos(3 \cdot \phi) + 0.118 \cdot \cos(5 \cdot \phi) = \dots [m/degree] \quad (11)$$

4 Schéma cinématique

4.1 Vecteurs positions

origine : centre de rotation verticale se trouvant sous les pâles principales.

position des pâles principales (pp) : vecteur verticale

position de l'hélice arrière : vecteur allant de l'origine vers l'hélice (h) arrière.

5 Angular momentum

5.1 Formula

$$\vec{L} = \vec{OA} \otimes \vec{P} = \vec{r} \otimes \vec{P} = \vec{r} \otimes m \cdot \vec{v} = \vec{I} \otimes \vec{\omega} \quad (12)$$

$\vec{L}$: Angular Momentum [$kg \cdot \frac{m^2}{s}$]

$\vec{OA}$ and r : position of the mass [m] according to a reference

$\vec{P}$: linear momentum [$kg \cdot \frac{m}{s}$]¹

$\vec{v}$: velocity [$\frac{m}{s}$] I : moment of inertia [$m^2 \cdot kg$]

ω : angular speed [$\frac{rad}{s}$]

Torque :

$$M = \frac{d\vec{L}}{dt} = \frac{d(\vec{I} \otimes \vec{\omega})}{dt} \quad (13)$$

if we consider a particule of mass m , $\vec{r}$ is the position of the center of mass. If it is a solid object, L is first computed according to the axis of rotation of the object :

$$\vec{L}_{ar} = \vec{I}_{ar} \otimes \vec{\omega}_{ar} \quad (14)$$

To compute the angular moment according to an other axis of rotation (new reference), we use the Huygens-Steiner theorem (or the Parallel axis theorem) :

$$\vec{L}_0 = \vec{I}_0 \otimes \vec{\omega}_{cm} \quad (15)$$

$$\vec{I}_0 = \vec{I}_{ar} + m \cdot d^2 \quad (16)$$

with d the distance between the axis of rotation of the object and the new reference.

5.2 Condition of stability

Main rotor(s):

$$\vec{L}_{mr} = \vec{r}_{mr} \otimes m_{mr} \cdot \vec{v}_{mr} = \vec{I}_{mr} \otimes \vec{\omega}_{mr} \quad (17)$$

Rear rotor :

$$\vec{L}_{rr} = \vec{r}_{rr} \otimes m_{rr} \cdot \vec{v}_{rr} = \vec{I}_{rr} \otimes \vec{\omega}_{rr} \quad (18)$$

assurer la stabilité lors du vol: les moments cinétiques doivent s'annuler. (poser la formule et résoudre)

$$\vec{L}_{mr} = \vec{L}_{rr} \quad (19)$$

or

The generated torque is compensated :

$$\sum \vec{M}_{mr} = \sum \vec{M}_{rr} \quad (20)$$

find a relation between ω_{mr} and ω_{rr} -> determine the transmission ratio

¹ $\vec{L}$ is perpendicular to both $\vec{P}$ and $\vec{r}$

5.3 Pivots à droite et à gauche

pour tourner à gauche ou droite, on ne doit plus satisfaire la condition de stabilité. le pilote utiliser le pédalier pour accélérer/ralentir l'hélice arrière. ainsi les moments cinétiques ne sont plus égaux.

calculer l'effet de rotation sur l'hélicoptère si l'hélice est accélérée/ralentie de 10,20,30,..%. mettre un tableau. calculer la vitesse de rotation dans ces cas-là.

$$\begin{bmatrix} 0 \\ 0 \\ l_1 \end{bmatrix}_{R_1} \vec{AB}_{R_2} = \begin{bmatrix} 0 \\ l_2 \\ 0 \end{bmatrix}_{R_2} \vec{BC}_{R_3} = \begin{bmatrix} l_3 \\ 0 \\ 0 \end{bmatrix}_{R_3} \vec{CD}_{R_4} = \begin{bmatrix} 0 \\ 0 \\ -l_4 \end{bmatrix}_{R_4} \quad (21)$$

.

.

.

$$\begin{aligned} \vec{OE}_R &= \vec{OA}_R + \vec{AB}_R + \vec{BB}_{1R} + \vec{B_1C}_{1R} + \vec{C_1C}_R \\ &\quad + \vec{CC}_{2R} + \vec{C_2D}_R + \vec{DD}_{1R} + \vec{D_1E}_R \end{aligned} \quad (22)$$

$$\begin{aligned} \vec{OF}_R &= \vec{OA}_R + \vec{AB}_R + \vec{BB}_{1R} + \vec{B_1C}_{1R} + \vec{C_1C}_R \\ &\quad + \vec{CC}_{2R} + \vec{C_2D}_R + \vec{DD}_{1R} + \vec{D_1E}_R + \vec{EF}_R \end{aligned} \quad (23)$$

$$\begin{aligned} \vec{OG}_R &= \vec{OA}_R + \vec{AB}_R + \vec{BB}_{1R} + \vec{B_1C}_{1R} + \vec{C_1C}_R + \vec{CC}_{2R} \\ &\quad + \vec{C_2D}_R + \vec{DD}_{1R} + \vec{D_1E}_R + \vec{EF}_R + \vec{FF}_{3R} + \vec{F_3G}_R \end{aligned} \quad (24)$$

$$\begin{aligned} \vec{OH}_R &= \vec{OA}_R + \vec{AB}_R + \vec{BB}_{1R} + \vec{B_1C}_{1R} + \vec{C_1C}_R + \vec{CC}_{2R} + \vec{C_2D}_R \\ &\quad + \vec{DD}_{1R} + \vec{D_1E}_R + \vec{EF}_R + \vec{FF}_{3R} + \vec{F_3G}_R + \vec{GH}_R \end{aligned} \quad (25)$$

6 Conclusion